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# CHALLENGES_AND_LEARNINGS.pdf
## Introduction
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During the implementation of this project, several challenges were encountered while working with Terraform, Jenkins, Docker, ECS, and AWS services.

Resolving these issues provided valuable troubleshooting experience and improved understanding of DevOps practices.

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## Challenge 1 - Jenkins Executor Offline
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### Error
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Waiting for next available executor.

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### Root Cause
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The Jenkins node became unavailable because available disk space was below the configured threshold.

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### Resolution
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- * Cleaned temporary files.

- * Released disk space.

- * Brought the Jenkins node back online.

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### Learning
```

Always monitor Jenkins node health and storage usage.

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## Challenge 2 - terraform.tfvars File Missing
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### Error
```

Failed to read variables file.

terraform.tfvars does not exist.

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### Root Cause
```

The Terraform pipeline was executed from an incorrect directory.

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### Resolution
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Verified the workspace structure using:

- * pwd

- * ls -la

- * find .

Updated the pipeline path.

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### Learning
```

Always verify the current working directory before running Terraform commands.

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## Challenge 3 - No Configuration Files
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### Error
```

No configuration files found.

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### Root Cause
```

The folder name in Jenkins did not match the actual repository structure.

Terraform expected:

Terraform

Actual folder:

Terrafrom

```
### Resolution
```

Corrected the folder path inside the Jenkins pipeline.

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### Learning
```

Naming consistency is critical in automation.

Challenge 4 - CloudWatch Module Error

Error

Unreadable module directory.

Root Cause

Terraform expected:

CloudWatch

Actual folder name:

Cloudwatch

Resolution

Updated the module source path.

Learning

Terraform module paths are case-sensitive.

Challenge 5 - AWS Credentials Error

Error

No valid credential sources found.

Root Cause

Terraform backend could not access AWS credentials.

Resolution

Configured AWS credentials within Jenkins.

Learning

Terraform backend authentication must be configured before initialization.

Challenge 6 - RDS Subnet Group Error

Error

Invalid subnet group name.

Root Cause

Resource naming configuration was invalid.

Resolution

Corrected subnet group naming values.

Learning

AWS naming conventions must always be followed.

Challenge 7 - Docker Login Failure

Error

400 Bad Request during ECR login.

Root Cause

The ECR repository URL was used instead of the ECR registry URL.

Resolution

Updated the Docker login command with the correct ECR registry endpoint.

Learning

Understand the difference between ECR Registry and ECR Repository.

Challenge 8 - ECS Image Pull Failure

Error

CannotPullContainerError

Image latest not found.

Root Cause

The application image was pushed using build numbers instead of the latest tag.

Resolution

Updated the Jenkins pipeline to push:
8byte-app:latest

Learning

A consistent image tagging strategy is important for deployments.

Challenge 9 - Application Not Accessible

Root Cause Investigation

Verified:

- * ECS Service
- * ECS Tasks
- * Security Groups
- * Target Group
- * Application Load Balancer
- * CloudWatch Logs

Resolution

Validated networking, target group configuration, and container port mapping.

Learning

Application troubleshooting should follow a structured approach from Load Balancer to Container.

<http://8byte-alb-2116250128.ap-south-1.elb.amazonaws.com>

Key Learnings

This project provided hands-on experience with:

- * Terraform
- * Jenkins
- * Docker
- * Amazon ECS Fargate
- * Amazon ECR
- * Amazon RDS PostgreSQL
- * CloudWatch
- * Application Load Balancer
- * Infrastructure as Code
- * CI/CD Automation
- * AWS Networking
- * DevOps Troubleshooting